

Lab 10

Lab Report II

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PHYS 1110

In lab 10, we employed two different methods of data collection and equations to extract a spring constant for a spring provided to us in lab. To do this, we first were tasked to add various masses to a hanger and pulled the system down about five to six centimeters and released; causing the spring to oscillate up and down. From these oscillations, we were able to use a stopwatch to time each oscillation and extract a period from it; which would be used to find the spring constant. The goal was to show that either method of collection and equations would lead to a similar, accurate spring constant.

Through the experiments, we were tasked with proving that both approaches would result in a accurate spring constant. Through data collection and the use of various equations, we attempted to calculate a spring constant and percentage error to ultimately determine which method yielded a more accurate result.

The method of data collection was very simple. Firstly, we only used the provided spring, no hanger and no masses present. We measured the distance from top of the spring to bottom. Then, we repeated this measurement with the provided hanger and subsequently 5 different masses. Once we had six total distances, we subtracted them from our initial measurement to extract the displacement in the spring caused by our hanger and mass systems.

Secondly, we were tasked with extracting out a period of oscillation for each of the different hanger and mass systems, not including the spring by itself. To do this, we took each system and pulled it down around five to six centimeters. When released, the system would oscillate up and down. Employing a stopwatch and ignoring the first oscillation to allow the system to stabilize, we timed how long it took for each system to oscillate at total of ten times; and recorded our findings. In total we had 6 data points for each of the predetermined masses.

Once all of our data was collected, we cleaned up the numbers by converting them to proper units. Our spring's displacement was notified as "x" which we had to convert to meters as was previously illuded. Our initial distance, from top of the spring to bottom with no attached hanger or masses was 0.08 meters. As for the hanger and mass systems, we carefully weighed them and converted them to kilograms, which were further multiplied by 10 meters per second for gravity to find weight, as is necessary for later equations. We then compiled this displacement and weight data into Table 1.

For our second set of collected data, we started off with just our 0.05 kilogram hanger. From there we added on the various predetermined masses and recorded their total weight. We then compiled our 10 oscillation periods, measured in seconds, and added everything into Table 2.

TABLE 1

TABLE 2

x (meters)	mg (kilograms * gravity)		10T (seconds)	mg (kilograms * gravity)
0.31	553		8	0.50
0.38	1053		9	0.60
0.54	1553		10	0.70
0.70	2053		11	0.90
0.86	2553		11.6	1.00
0.95	2825		12.7	1.20

Once all of our data was compiled, we could begin our analysis. In order to achieve our stated goal of extracting a spring constant, we employed the use of two different equations which are presented in Table 3. Additionally, we graphed each data set using a scatter plot and trendline within Microsoft Excel which would allow us to extract our two spring constants.

TABLE 3

EQUATION 1

EQUATION 2

$m * g = k * x$	$m = (k / (4\pi^2)) * T^2$
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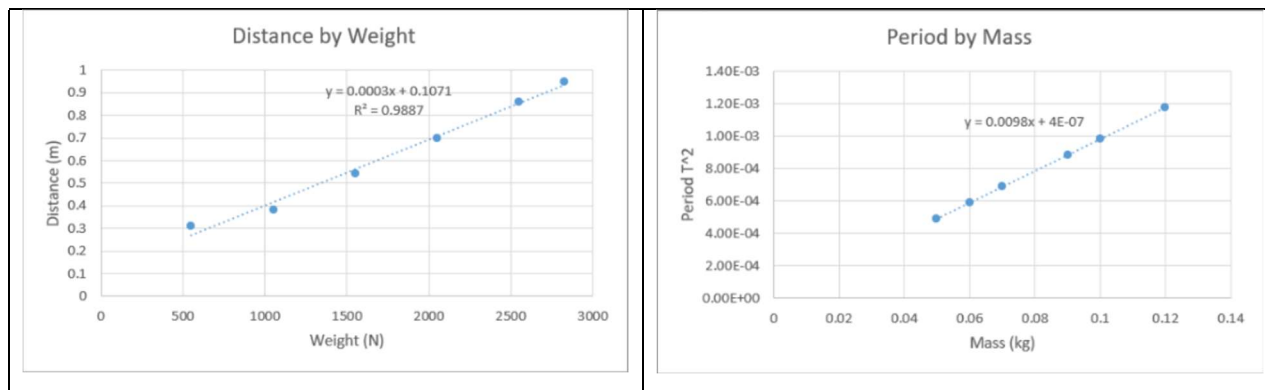
For the first set of data we used Equation 1. As previously mentioned, the weight and displacement points were graphed against each other which then allowed us to run a linear fit trendline through the data, giving us our first spring constant of 3.3 Newtons/Meters. The graph, for reference is provided in Table 4.

For our second set of data we used our second equation from Table 3. Once again, we were able to extract the spring constant from this data set using a linear fit trendline using our masses against our periods squared on a scatter plot. To get our periods squared, we simply divided our initial periods from Table 2 and then squared them to get the data in the form needed by Equation 2. Once completed, and we took the slope of the linear fit and multiplied it by $4\pi^2$ per Equation 2 to finally extract our second spring constant of 2.9 Newtons/Meters.

TABLE 4

Displacement (Data set one)

Oscillation (Data set two)



Once the equations portion was completed, we were left with two spring constants: 3.3 Newtons/Meters and 2.9 Newtons/Meters. From this, we simply calculated a standard deviation of 0.29 using Excel's native STD function and a percentage error of around 13%. Doing this allows us to quantify the quality of our extracted spring constants and how the difference between the two methods.

When comparing both spring constants, it is clear that both methods yielded similar results. Our methods of data collection and subsequently our equations and graphs were largely successful and accurate. A visual representation of our data collection accuracy is found in our two graphs from Table 4. All data points were very close to our trendline, meaning that the correlation between our two data set, for both methods, were largely accurate and led us to not having any problems in data collection or analysis. Similarly, our standard deviation and percentage error were small enough for me to conclude that both methods would result in similar spring constants, which was in fact the case.

One major contributing factor to the small but present difference in magnitudes can be blamed on human error. This type of error can always be minimized but never completely eliminated and it shows that even when we carefully gathered our data, mistakes in reaction time and estimation were still present. In this experiment we had many chances for human error to play a part in data collection. At every measurement, we were solely reliant on our eyes for distances and our reaction times as we were estimating the period of oscillations with our eyes and a stopwatch.

I believe the first spring constant, extracted using Equation 1 to be more accurate. This is most likely due to the room left to human error, which was significantly larger in the second version of data collection. In the first version, our only significant room for error was determined by how accurately we could measure distance / displacement of the spring and hanger mass system using a ruler. I believe we did this to the very best of our abilities, minimizing human error and ultimately providing us with a more accurate spring constant.

As for our second version, the room for error was more significant, as previously stated. When pulling the hanger mass system down, we did not measure how far we were pulling down, which would obviously introduce discrepancies in our data as a larger initial displacement would undoubtedly affect our oscillations. In addition, we relied heavily on our reaction times to gather the periods of oscillation. Where each oscillation stopped and new one began were determined entirely on our ability to react to the system's movements and subsequently stop the stopwatch when ten full oscillations were perceived. It is clear that the second version of data collection introduced significantly more chances for human error to effect our final extracted spring constant.

In conclusion, we employed two different methods of data collection and analysis using varying equations to extract out our spring constants. Even though our second method no doubt invited more human error, our spring constants remained similar, which is a testament to our data collection accuracy and the viability of both methods. Ultimately, our first version, which employed Equation 1, was the more accurate spring constant at 3.3 Newtons/Meters as it had less human error involved.